

Video2Entities: A computer vision-based entity extraction framework for updating the architecture, engineering and construction industry knowledge graphs

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ABSTRACT

Due to the decentralisation and complexity of knowledge in the architecture, engineering and construction (AEC) industry, the research on knowledge graphs (KGs) is still insufficient, and most of the research focuses on text-based KG structuring or updating. Entity extraction, a sub-task of knowledge extraction, is critical in general KG update approaches. While the mainstream approach for this task generally uses textual data, visual data is more readily available, more accurate and has a shorter update cycle than textual data. Therefore, this paper integrates zero-shot learning techniques with general KGs to present a novel framework called “video2entities” that can extract entities from videos to update the AEC KG. The framework combines the perceptual capabilities of computer vision with the cognitive capabilities of KG to improve the accuracy and timeliness of KG updates. Experimental results demonstrate that the framework can extract “new entities” from architectural decoration videos for AEC KG updates.

1. Introduction

In many industrial sectors, knowledge is considered an essential resource [1] and effective knowledge management is vital to their competitiveness [2]. Although knowledge management is receiving increasing attention in the architecture, engineering and construction (AEC) industry [3], the project-oriented, experience-specific, context-specific and decentralised nature of knowledge [4] still limit the acquisition and reuse of knowledge in this industry. Knowledge graph (KG) is an efficient tool for knowledge management that can effectively reuse, retrieve and visualise knowledge in a structured way. Since Google first introduced the concept of KG in 2012, KG has become popular in the industry and academia, and many areas such as semantic search [5], intelligent Q&A [6] and personalised recommendation systems [7] have benefited from KG technologies.

The KG, an organic combination of a knowledge base and an ontology, consists of a data layer and a schema layer, as shown in Fig. 1. The data layer is mainly composed of a knowledge base of entities, attributes and relationships. The schema layer built on top of the data

layer standardises knowledge expression in the data layer through ontologies. The ontology is the conceptual template that defines concepts, conceptual attributes and conceptual relationships. The knowledge base formed by ontology has a hierarchical structure with low redundancy. Besides, depending on the area covered, one can divide KGs into the domain KG and the general KG. Domain KGs contain domain-specific data with diverse attributes and data patterns. Compared with domain KGs, general KGs focus on the breadth of knowledge, covering more domains and integrating more entities, while their knowledge is not in-depth and precise.

Researchers have presented a great deal of work on ontology in many aspects of the AEC field. For example, Park et al. [4] focused on capturing and identifying construction knowledge characteristics and applying these characteristics to develop an ontology-based construction knowledge retrieval system. Niu and Issa [8] presented a delay analysis ontology for claim management. Ma et al. [9] aimed to formalise specifications for construction cost estimation via ontologies. By establishing ontology-based frameworks, Zhong et al. [10] identified potential hazards from images, Li et al. [11] performed structural health

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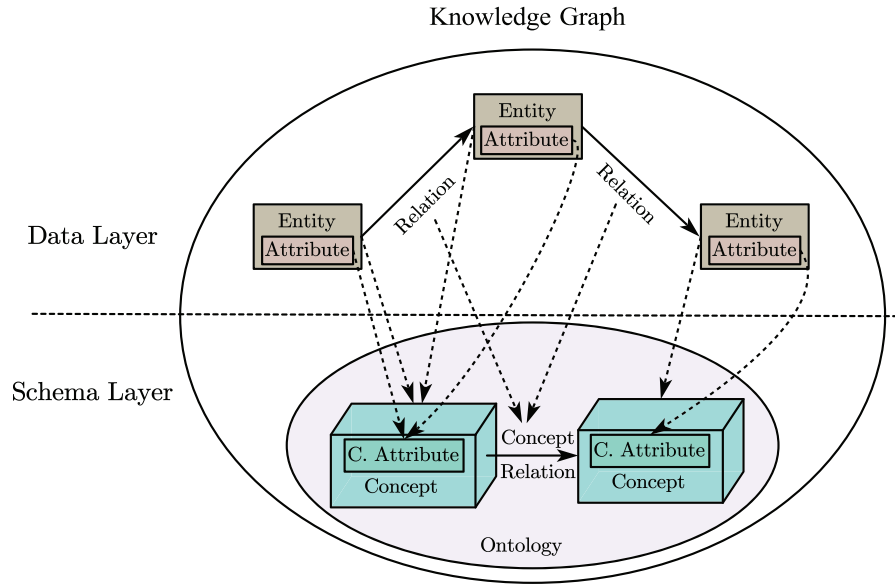


Fig. 1. The logical structure of the knowledge graph.

monitoring (SHM) for fine-grained bridge structures, and Chen et al. [12] performed facility maintenance management (FMM). However, due to the complexity of AEC KGs, little research has been devoted to KG in this industry. Rasmussen et al. [13] applied KG to manage interrelated project information. Fang et al. [14] established a risk identification system that combines object detection (OD) and KG. Zhu et al. [15] constructed a KG of geographic information.

Nevertheless, none of these researchers has investigated KG updates or vision-based knowledge extraction. In particular, the KG update is a continuing problem when structuring KGs. Typically, there are two methods to update KG: full and incremental updates [16]. Full updates mean building the KG from scratch with all the updated data as input. Incremental updates fuse new knowledge in the existing KG using the current new data as input. While the incremental update consumes fewer resources than the full update, it still requires significant manual intervention (e.g., defining rules) [17]. Therefore, this paper proposes a semi-automated framework to reduce manual intervention when incrementally updating AEC domain-specific knowledge to the general KG data layer.

Entity extraction, a subtask of knowledge extraction, is one of the most critical tasks. Although previous studies have extensively used textual data for this task, this paper chooses video data as the primary source of knowledge for the following reasons. First, vision is the most convenient and effective way to perceive information. Most of the information that humans acquire comes from vision. Second, video data contains richer information than text data, helping to reduce deception and ambiguity. Third, video data is updated much faster than text data, making KG updates more efficient. Moreover, as deep learning (DL) has evolved, computer vision has made significant breakthroughs in image classification and detection since 2012 [18,19], bringing it closer to human vision. As a result, one can extract entities from videos using computer vision algorithms and update them into the KG.

However, not all computer vision algorithms are suitable for entity extraction. In theory, general supervised learning approaches can extract entities as long as there are sufficient training instances for all classes. Due to the increasing number of new categories in the AEC domain and the long-tailed data distribution, it is almost impossible to access the tail (or rare) classes' training instances. As supervised learning approaches cannot transfer known knowledge to unknown knowledge, they cannot recognise "new entities (entities outside the training set)". Although few-shot learning can identify new categories (unseen classes) with limited supporting samples, it cannot recognise

new categories in the absence of supporting samples. Therefore, we need to find a new technique that can recognise "new entities" using a limited quantity of seen class training data, which is the main focus of zero-shot learning (ZSL), aiming to classify unseen class instances [20].

Motivated by the above observations, we propose a novel framework called video2entities, which aims to update the AEC domain KG by extracting entities from the video using ZSL techniques. The framework consists of two iterative processes: a forward process and a backward process. The forward process initialises the video entity detector (VED) module using word embeddings as prior information to train the zero-shot action recognition (ZSAR) and zero-shot object detection (ZSD) models. This module can recognise seen and unseen entities. Due to the absence of well-established domain KGs in the AEC field, the forward process updates the extracted entities based on general KGs. The backward process combines the updated general KG with word embeddings to extract semantic embeddings using a graph convolution network (GCN). In the next iteration, the forward process takes advantage of the better VED built by the integration of semantic embeddings and extracts entities. Our framework combines the perceptual capabilities of computer vision with the cognitive abilities of KG to iteratively improve the performance of the VED. As a result, we can extract most of the seen and unseen entities through multiple iterations.

This paper addresses the video entity extraction problem of AEC domain KG update by proposing a novel framework that can extract unseen entities from videos. Specifically, this paper addresses the unseen entities recognition problem by integrating the ZSL technique with general KG. This paper first introduces the ZSL technique into the video entity extraction task to the best of our knowledge. Besides, this paper has performed comprehensive experiments on architectural decoration videos. The results show that the framework can extract the seen and unseen entities from the videos to update AEC domain KGs.

2. Related works

2.1. Entity extraction

Entity extraction, also known as named entity recognition (NER), aims to extract entities from text. Entity extraction is the basis of solving natural language processing (NLP) problems, and it is also the most fundamental task in knowledge extraction. Methods of extracting entities from textual data fall into rule-based methods [21], statistical model-based methods [22–25] and deep learning-based methods

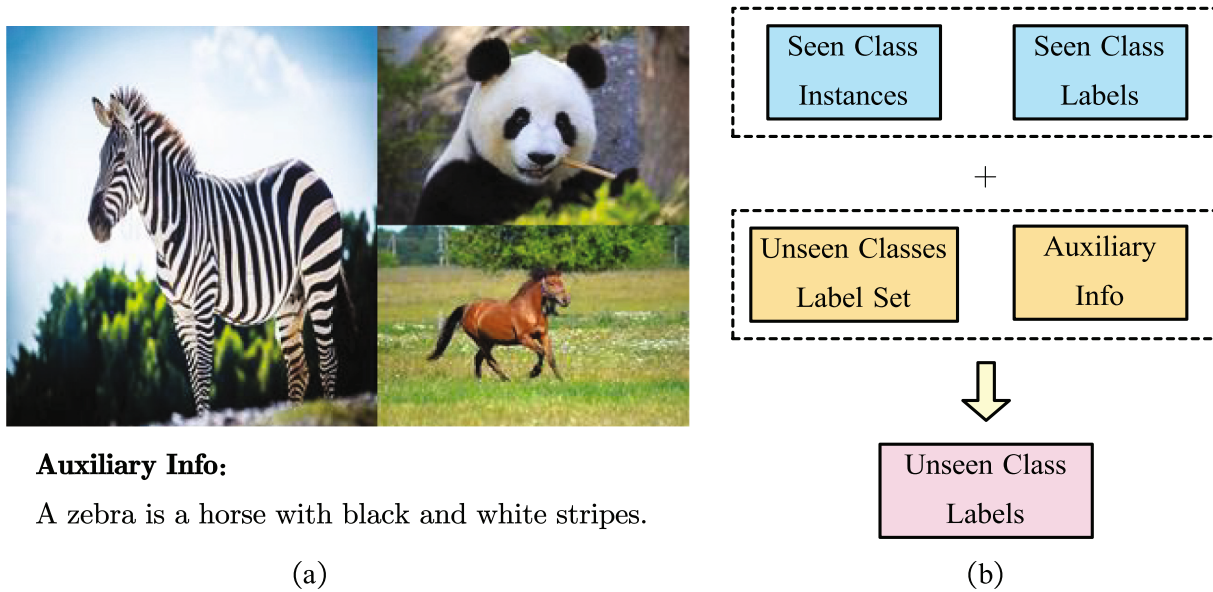


Fig. 2. Schematic diagram of the principle of ZSL.

[26–28].

Rule-based methods require linguists to manually construct rule templates and select specific features, including statistical information, punctuation, lexicality, directional words and word headings. However, rule-based methods are cumbersome and laborious, with long update cycles and poor portability. The statistical model-based methods regard NER as a sequence labelling problem. The predicted tags in sequence labelling problems are related to the current input features and the previously predicted tags compared with classification problems. The main probabilistic graph models are the hidden Markov model (HMM) [25], maximum entropy model [23], support vector machine (SVM) [24] and conditional random field (CRF) [22]. With the widespread application of DL in NLP, deep neural networks have been successfully applied to NER problems. Compared with statistical model-based methods, deep learning-based methods directly take word embeddings as input and implement end-to-end NER through neural networks, such as LSTM-CRF [27], LSTM-CNNs-CRF [26], and attention-based models [28].

2.2. Zero-shot learning (ZSL)

The nature of vision-based KG updates requires that computer vision methods should recognise “new entities”. Although traditional supervised learning or few-shot learning methods cannot accomplish this task, ZSL techniques can introduce additional auxiliary information or prior knowledge to complete this task. In this study, word embeddings are used as prior knowledge as it is the vectorisation of the word that can measure the semantic distance between each word. As shown in Fig. 2 (a), with known knowledge (i.e., black and white panda and ordinary horse) and auxiliary information “A zebra is a horse with black and white stripes”, we know that the unseen instance label is zebra. As shown in Fig. 2(b), ZSL is trained using the instances and labels of seen classes, applying the unseen labels and extra auxiliary information for testing. In recent years, many tasks in computer vision, such as object detection (i.e., ZSD) [29] and action recognition (i.e., ZSAR) [30], have benefited from ZSL.

There are two main types of existing ZSL methods. One is the projection-based method, and the other is the similarity calculation method. The projection-based method computes the distance between the visual features mapped to the semantic space and the semantic embeddings to recognise the unseen classes [31,32]. The similarity calculation method uses a classifier trained on the seen classes in visual

space to determine the distance between the unseen and seen class instances in semantic space [33].

2.3. ZSL on entity extraction

ZSAR and ZSD methods can recognise construction processes or detect tools, materials and corresponding attributes for a construction video. Zero-shot image or video captioning (ZSIC/ZSVC) methods can also generate a textual description of images or videos, along with related entities and attributes. In theory, all of the above techniques can extract entities and attributes. However, while the ZSIC/ZSVC method can simultaneously generate textual descriptions of objects, scenes and actions, they require an additional entity extraction step from the generated text, and their data annotation efforts are more complicated than those of the ZSAR or ZSD method. Besides, ZSIC/ZSVC methods rely on the upstream ZSAR or ZSD model. Therefore, ZSAR and ZSD are more suitable than ZSIC/ZSVC for the KG update task.

Existing ZSAR methods are generally divided into four types based on different prior information: attribute [34,35], word embedding [36–38], KG [30] and generative adversarial networks (GAN) [39,40]. In early ZSAR works, experts manually defined and labelled attributes of actions as prior information. However, handcrafted attributes are unsuitable, more subject to labelled data and sensitive to incorrect annotations. As a result, researchers have turned to word embedding. For example, Xu et al. [37,38] embed labels and videos using word embeddings as a shared semantic space. Subsequently, methods using semantic relations such as KG as prior knowledge have shown better performance. For example, Gao et al. [30] first presented a GCN-based ZSAR model to extract semantic information from KGs, and the performance of this model was state of the art at that time, which suggested that the prior information of KG is more potent than that of the attribute and the word embedding. The difficulty with ZSL is the lack of data on unseen categories. If unseen categories were available, ZSL could be easily converted to supervised learning, so some scholars have used GAN to “generate” data on unseen categories [39,40].

Existing ZSD methods also fall into four categories: attributes [29], word embedding [41], text description [42] and GAN [43,44]. The earliest work on ZSD, proposed by Rahman et al. [41], assumed that all the foreground candidate boxes in an image could be obtained through a region proposal network or edge box algorithm [45], thus reducing the detection task to a classification task. In contrast to work [41], Zhu et al. [29] argued that general object detection methods treated the unseen

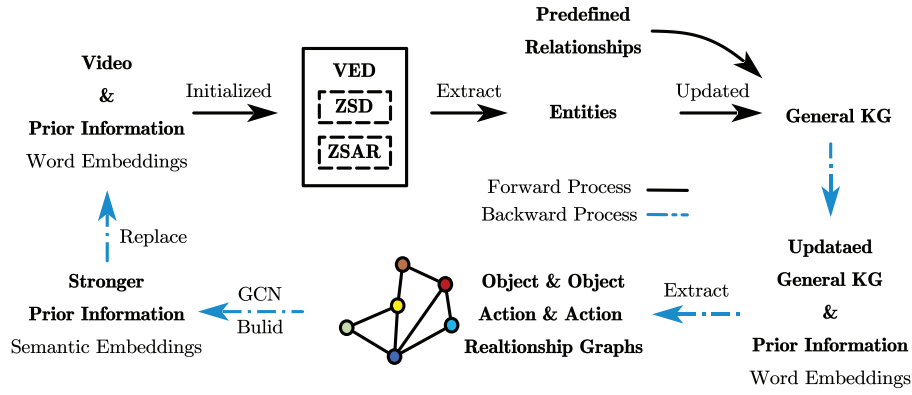


Fig. 3. Schematic diagram of the Video2Entities framework.

class as a background class, resulting in a lower confidence score for the unseen class. Li et al. [42] chose text descriptions instead of word embeddings as prior information. Works [43,44] used GAN to generate unseen classes' visual features, transforming the ZSL task into a supervised classification learning task.

Considering the superiority of ZSD and ZSAR over ZSIC/ZSVC, the VED module in the framework builds on the foundation of ZSD and ZSAR. Additionally, since KG has more robust prior information than attribute and word embedding, the framework integrates the relationship graphs extracted from KG and word embedding as prior information to optimise the VED module.

3. Video2Entities

Our framework updates the AEC domain KG with predefined relationships and entities extracted by the VED module. The VED module comprises ZSAR and ZSD models, where the ZSAR model recognises construction processes, and the ZSD model detects construction materials and tools. As shown in Fig. 3, the framework contains two processes: the solid black line shows the forward process, and the dash-dot blue line shows the backward process. The forward process initialises the VED module using word embeddings as prior information. Following the VED entity extraction, the forward process updates the extracted

entities in the general KG with the predefined relationships. By integrating the updated general KG with the corresponding word embedding, the backward process can structure action-to-action and object-to-object relationship graphs. Afterwards, GCN extracts the semantic information in the relationship graph to construct semantic embeddings. In the next iteration, the semantic embedding replaces the word embedding in the forward process.

3.1. Problem definition

The VED module, which contains the ZSAR and ZSD models, is the critical module in our study. Formal definitions of the VED, ZSAR and ZSD are given below.

3.1.1. VED

Formally, given a dataset $D = \{Input^{s+u}, X^{s+u}, Y^{s+u}\}$ with $(S+U)$ entities, where $Input$ represents the input videos or images, X represents the prior information, Y represents the labels of $Input$, and S and U represent the seen and unseen entities, respectively. In the semi-supervised setting, where the ground truth labels of the seen entities Y^s and the prior information X^{s+u} of both the seen and unseen entities are known, the goal is to infer the unseen labels Y^u from the unseen entities U . The function $F(\cdot) = \{f_{ZSAR}(\cdot), f_{ZSD}(\cdot)\}$ denotes the VED module with

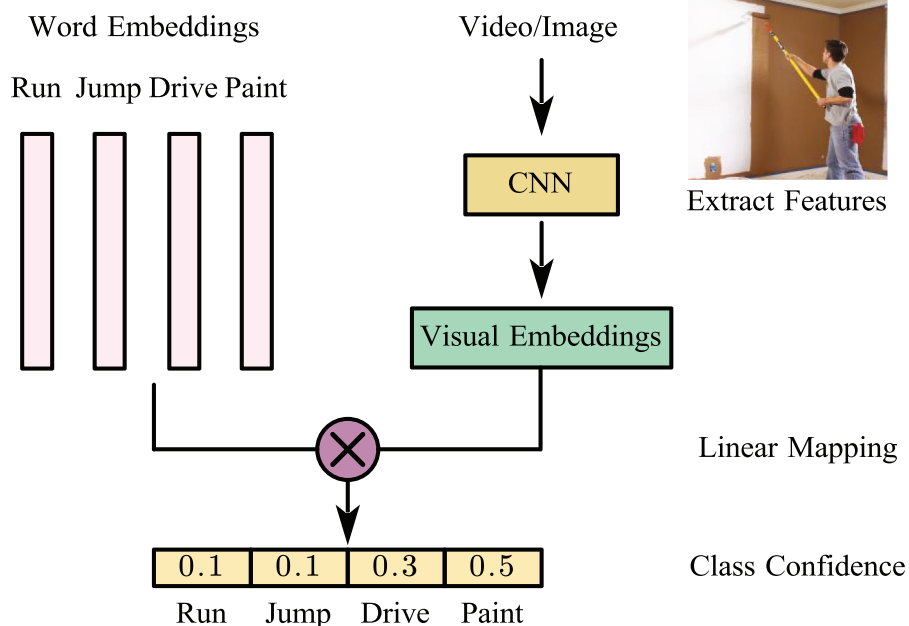


Fig. 4. Diagram of initialising VED in the forward process.

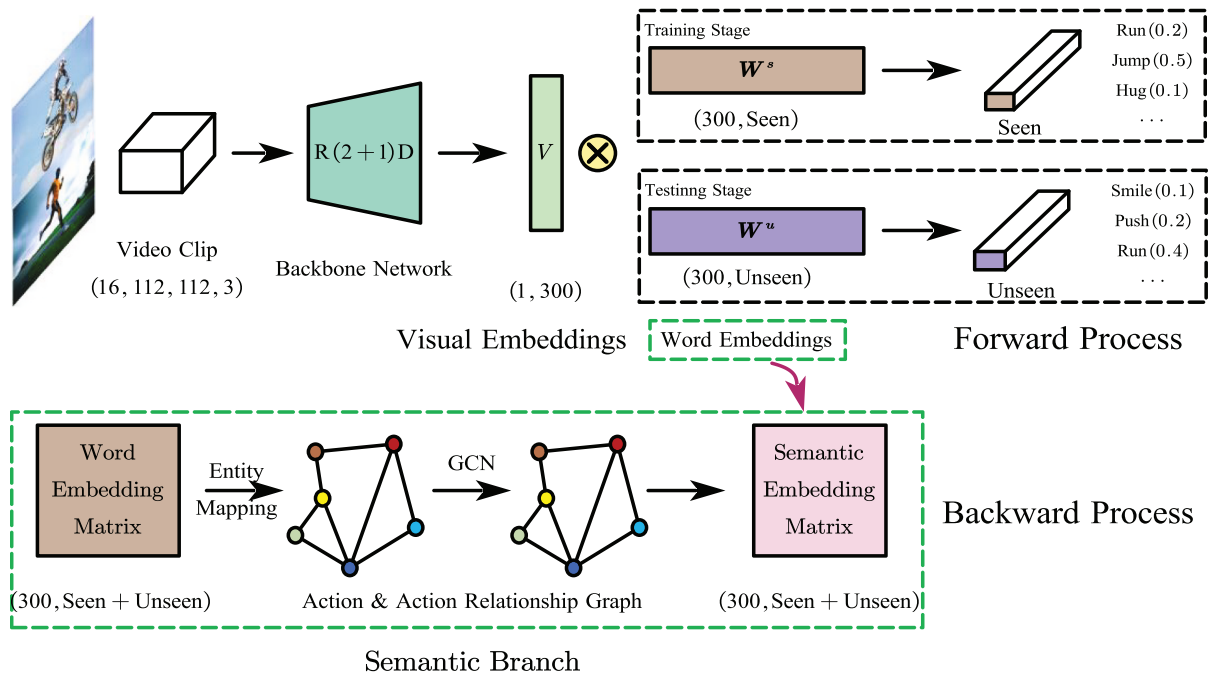


Fig. 5. A ZSAR model built on word embeddings or semantic embeddings.

two functions $f_{\text{ZSAR}}(\cdot)$ and $f_{\text{ZSD}}(\cdot)$, representing the ZSAR and ZSD models, respectively.

3.1.2. ZSAR

Assume there is a video training dataset $D_v^s = \{V^s, Y_v^s\}$ containing seen category videos V^s and action labels Y_v^s , where each video instance $v^s \in V^s$

is associated with an action label $y_v^s \in Y_v^s$. Likewise, assume there is a video test dataset $D_v^u = \{V^u, Y_v^u\}$ containing unseen category videos V^u and action labels Y_v^u . Here, $Y_v^s \cup Y_v^u = Y$, $Y_v^s \cap Y_v^u = \emptyset$, where Y_v represents the collection of seen and unseen action labels. The goal of the ZSAR model $f_{\text{ZSAR}}(\cdot)$ is to minimise the objective function $L(D_v^s)$ and generalise it to D_v^u .

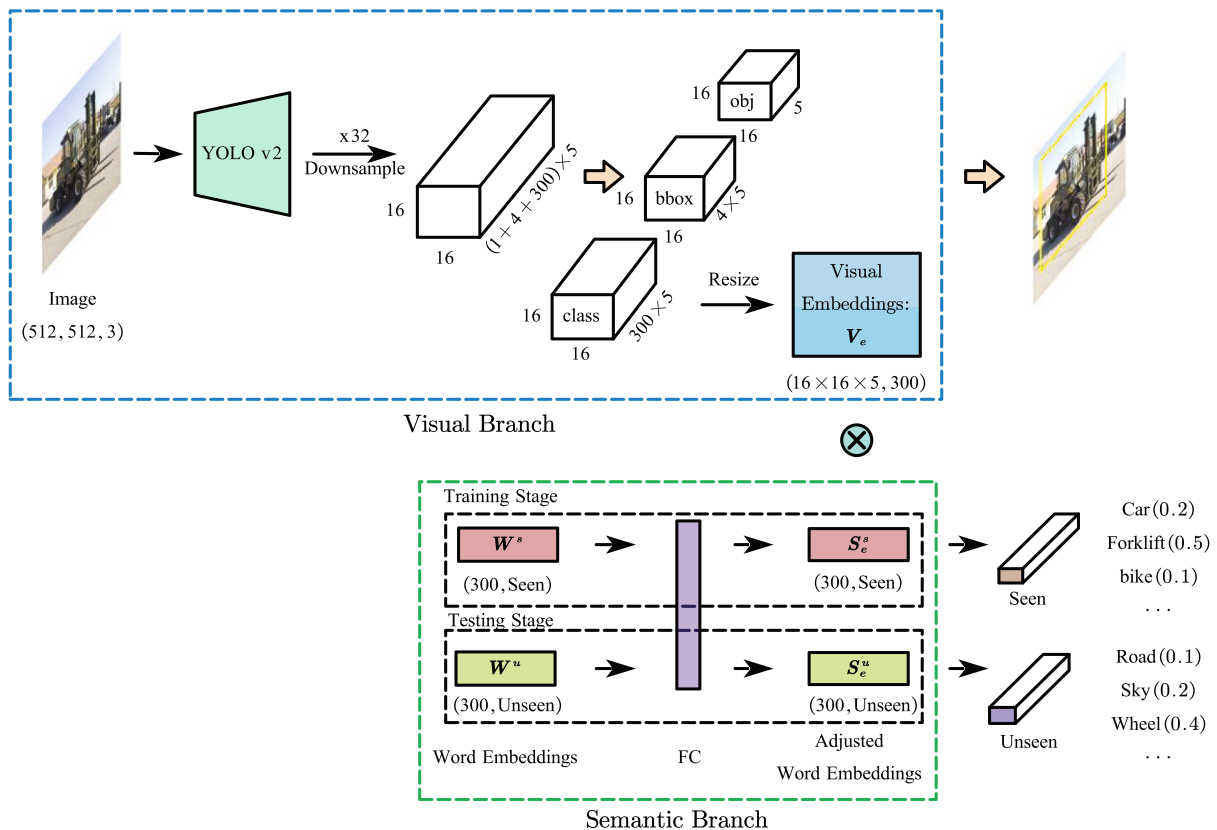


Fig. 6. A ZSD model based on word embeddings.

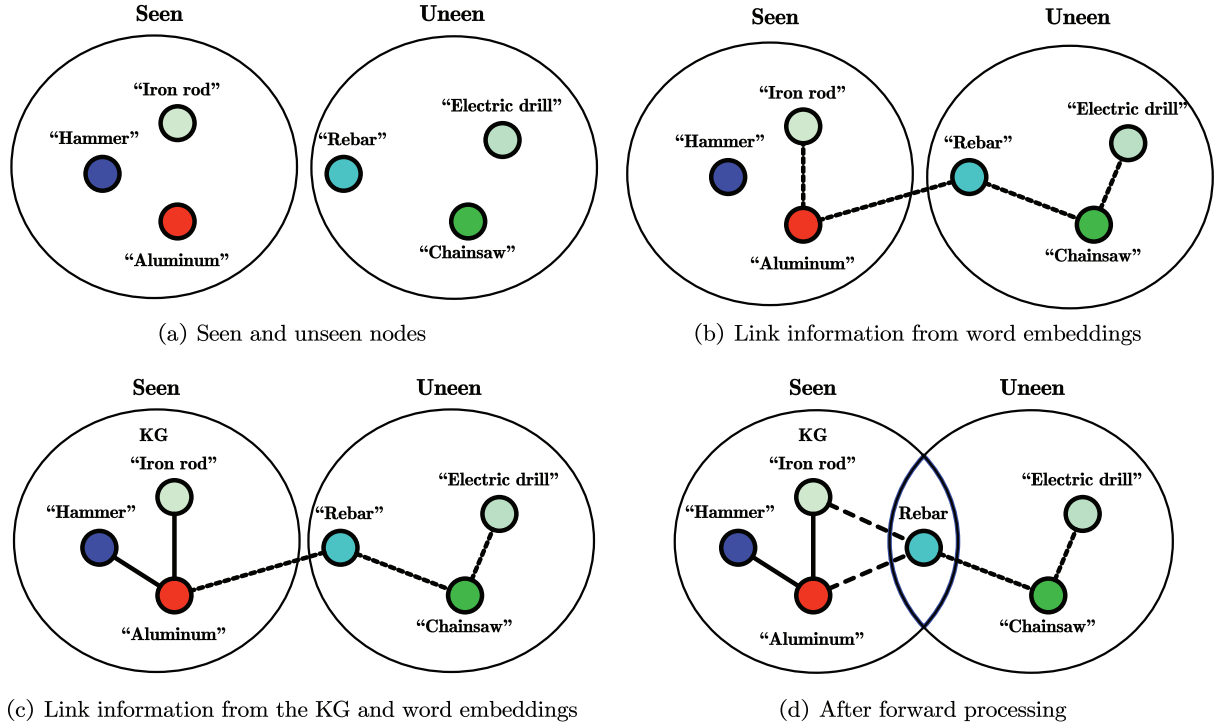


Fig. 7. Relationship graph.

3.1.3. ZSD

Assume there is an image training dataset $D_I^s = \{I^s, B_I^s, Y_I^s\}$ containing seen category images I^s and corresponding bounding boxes B_I^s and object labels Y_I^s , where each image bounding box instance $b_I^s \in B_I^s$ is associated with an object label $y_I^s \in Y_I^s$. Likewise, assume there is a test dataset $D_I^u = \{I^u, B_I^u, Y_I^u\}$ containing unseen category images I^u and corresponding bounding boxes B_I^u and unseen object labels Y_I^u . Here, $Y_I^s \cup Y_I^u = Y_I$, $Y_I^s \cap Y_I^u = \emptyset$, where Y_I denotes the collection of seen and unseen object labels. The goal of ZSD model $f_{ZSD}(\cdot)$ is to minimise the objective function $L'(D_I^s)$ and generalise it to D_I^u .

3.2. Forward process

The forward process's main idea comes from the literature [46], which linearly maps visual embeddings into the semantic space. As shown in Fig. 4, the left branch is the word embedding of all candidate labels, and the right branch is the visual feature extraction process using a convolutional neural network (CNN). We calculate class confidence using the inner product. The model's compatibility is 1 if the word embedding is compatible with the visual embedding; otherwise, it is 0. The class with the highest confidence is the label of the video or image.

3.2.1. Build the ZSAR model by word embeddings

The upper branch of Fig. 5 shows the structure of the word embedding-based ZSAR model, which leverages the R (2 + 1)D CNN [47] model as the backbone network to extract the video features, with the number of neurons in the last fully connected layer of the backbone network modified to 300. In this way, we can take a 16-frame video segment of size $112 \times 112 \times 3$ (H × W × C) as input and obtain a visual embedding V of length 300 through the backbone network. We calculate the inner product between the visual embedding V and the seen class word embeddings $W^s \in \mathbb{R}^{300 \times \text{seen}}$ in the training stage, and obtain the seen class confidence vector via the softmax function, as shown below

$$\text{Seen Class} = \text{softmax}(VW^s)$$

In the test stage, the unseen class word embeddings $W^u \in \mathbb{R}^{300 \times \text{unseen}}$

replace the seen class word embeddings W^s , i.e., the test phase is a ZSL task where the test data contains only unseen instances. In the generalised ZSL (GZSL) task, the seen and unseen class word embeddings $W \in \mathbb{R}^{300 \times (\text{seen} + \text{unseen})}$ replace the seen class word embeddings W^s , i.e., the test data contains both seen and unseen instances. Due to the class imbalance problem, the model's loss function adopts focal loss [48].

3.2.2. Build the ZSD model by word embeddings

As shown in Fig. 6, the word embedding-based ZSD model contains two branches: a visual branch and a semantic branch. Based on the one-stage object detection model YOLOv2 [49], the visual branch modifies the number of output channels of the YOLOv2 to 300. The visual branch aims to extract visual features and obtain the boundary box coordinates and each foreground object's confidence level. For an input image of size 512×512 , YOLOv2 yields a tensor output of size $16 \times 16 \times 1525$ after 32 downsamplings. This tensor contains three parts: an object confidence tensor (size $16 \times 16 \times 5$), a bounding box coordinate tensor (size $16 \times 16 \times 20$) and a visual embedding tensor (size $16 \times 16 \times 1500$). The visual embedding tensor is then resized to a two-dimensional matrix (size 1280×300), denoted as $V_e \in \mathbb{R}^{1280 \times 300}$. The semantic branch consists of a fully connected layer and a LeakyReLU [50] activation function. The semantic branch does not change the shape of the word embedding matrix, and the semantic embedding matrix is denoted as $S_e^s \in \mathbb{R}^{300 \times \text{seen}}$ in the training stage and $S_e^u \in \mathbb{R}^{300 \times \text{unseen}}$ in the testing stage. We still adopt linear mapping to align the visual and semantic spaces. After normalisation by the softmax function, we obtain the class confidence vectors. The loss functions for confidence, bounding box and class in the ZSD model are consistent with YOLOv3 [51].

3.2.3. Extract video entities and update KGs

After building the ZSAR and ZSD models by word embeddings, the VED module can extract video entities. The VED module will extract incorrect entities due to the word embedding-based ZSAR and ZSD models' performance limitations, therefore statistical analysis of the ZSAR and ZSD results is required. In a video, the construction processes, materials and tools are recognised every 16 frames, and the predicted

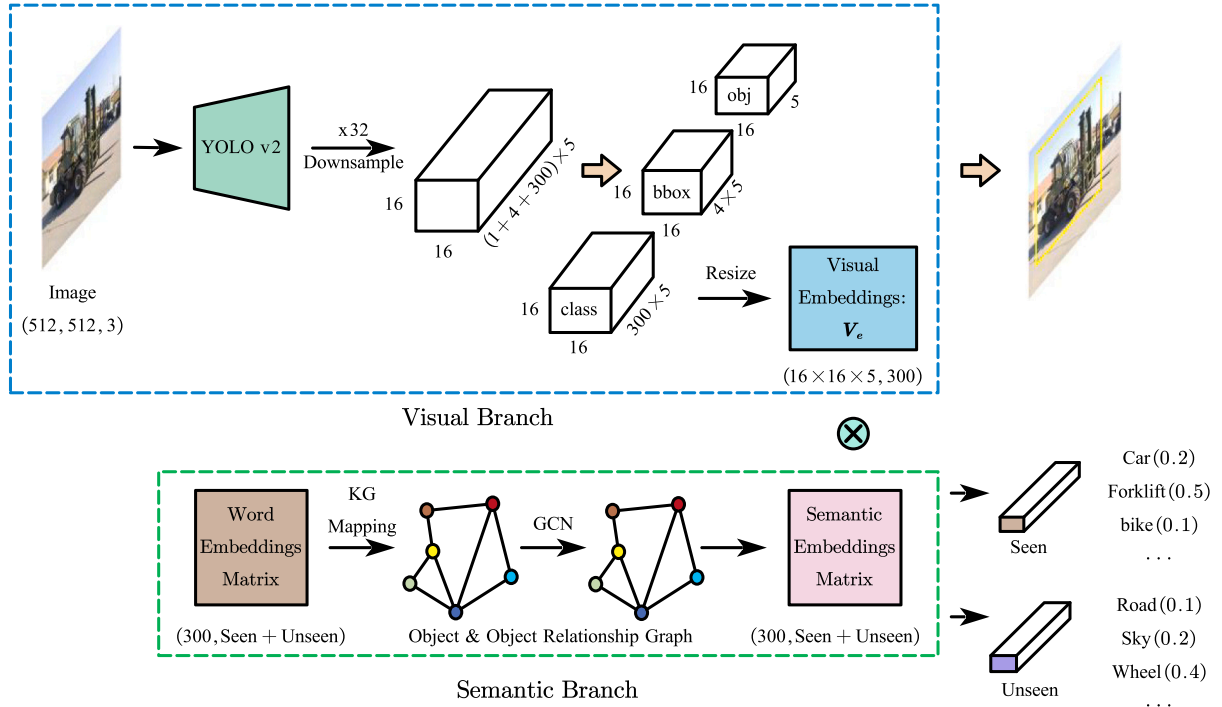


Fig. 8. A ZSD model based on semantic embeddings.

label with the most cumulative occurrences serves as the entity extracted in this video clip. In each video clip, the top-N materials or tools with the same construction process label are counted to prevent accidental errors. The top-N materials or tools are considered as the corresponding tool or material entities to that construction process. After multiple iterations, we can reduce the possibility of extracting incorrect entities.

Although a large-scale general KG tends to lack most of the AEC domain's entities and relationships, it still contains some entities relevant to the domain, such as concrete, chainsaws and electric drills. Therefore, general KGs such as ConceptNet 5.5 [52] were considered in this study as a basis for updating AEC domain-specific entities and relationships. In order to correlate discovered unseen entities with entities in existing KGs, we combine the entities in KGs and detected unseen entities (i.e. actions and objects) with predefined relationships (e.g., used for) into triplets of the form <action, predefined relationship, object>. These triplets can be easily stored in graph databases such as Neo4j [59], which manually or semi-automatically store and manage triplets as graphs. Thus, in Neo4j, detected entities can be associated with KG entities, as some entities in the triplets exist in the KG, meaning that we only update the KG data layer and not the ontology. However, it is also worth noting that exceptions may arise where both the action and the object are unknown, which inevitably requires manual intervention to determine the relationship between the new entity and those in the KG, as the newly added triplets have no connection to the existing KG.

3.3. Backward process

The prior information has a crucial impact on the performance of the ZSL model. The nature of ZSL determines that unknown entities need to be semantically related to known entities. If there are no or weak semantic connections between entities, e.g., some categories of entities with large sparsity, then identifying these categories of entities may be challenging, as they are close to being isolated at the semantic level. To reduce the effect of entity sparsity, prior information is needed to strengthen the semantic links between known and unknown entities. In particular, a reasonable KG can lead to more robust prior information

than word embedding. Therefore, the backward processing aims to enhance the prior information by distilling knowledge from the updated general KG and word embeddings, thus optimising the VED module.

3.3.1. Strengthen the prior information

Rather than using the large-scale KG directly, the backward process refines the KG's knowledge to construct a much smaller relationship graph containing only the semantic relationships between objects and objects or actions and actions. In a large-scale general KG such as ConceptNet 5.5 [52], entities belonging to the AEC domain are considered to be part of the seen classes, and the remaining seen classes are in the training set. Entities that neither exist in the training set nor the KG are unseen classes. We use word embeddings to link the unseen classes to the seen classes. First, word embeddings of all seen and unseen classes are collected, and then the cosine similarity is calculated in pairs. Two classes are linked together if their similarity is greater than a specified threshold. As shown in Fig. 7(b), "Iron rod", "Aluminium", "Electric drill", "Chainsaw" and "Rebar" are linked together. After the forward process, part of the unseen classes becomes seen classes, and all seen classes are updated to the general KG. Therefore, the link information between these unseen classes and all the seen classes in the KG is successfully acquired. In Fig. 7(d), "Rebar" is an unseen class. After the forward process, it becomes the seen class and acquires links to entities in KG. The final entity-to-entity (i.e. action-to-action and object-to-object) relationship graphs between the seen and unseen classes are derived by integrating the link information from word embedding and KG. Specifically, each node in the relationship graph is an entity, and two nodes are linked if they have a relationship in between, or if their cosine similarity is greater than a specific threshold.

If the prior information of the forward process is also seen as a relationship graph, then the link information of this graph only considers the word embedding, as shown in Fig. 7(b), while the link information of the backward process integrates both the word embedding and KG, as shown in Fig. 7(c). Therefore, it is reasonable to believe that the backward process's prior information is more robust than that of the forward process, and more robust prior information should lead to better

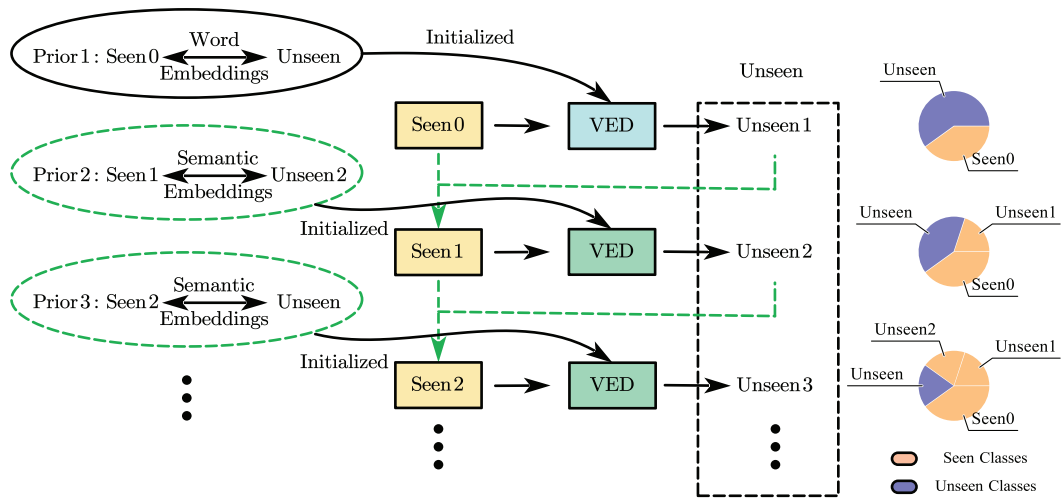


Fig. 9. The overall workflow of the video2entities framework.

performance.

3.3.2. Build the ZSAR model by action and action relationship graph

As shown in the lower branch of Fig. 5, the main difference between the relationship graph-based ZSAR model and the word embedding-based ZSAR model is the semantic branch. The semantic branch of the relationship graph-based ZSAR model extracts the semantic features from the relationship graph and word embeddings using GCN proposed in the literature [53]. Specifically, for a word embedding matrix of size $300 \times (\text{seen} + \text{unseen})$, a $300 \times (\text{seen} + \text{unseen})$ semantic embedding matrix can be obtained after three graph convolution layers, which means that all nodes (both seen and unseen) in the action and action relationship graph are used in the GCN, thus enabling the model to generalise the known knowledge to the unknown. The rest of the

relationship graph-based ZSAR model is consistent with the word embedding-based ZSAR model.

3.3.3. Build the ZSD model by object and object relationship graph

The structure diagram of the relationship graph-based ZSD model is shown in Fig. 8. The relationship graph-based ZSD model is adapted from the word embedding-based ZSD model with the semantic branch replaced. The model still leverages the idea of linear mapping to project visual features into the semantic space. The rest of the relationship graph-based ZSD model is the same as the word embedding-based ZSD model.

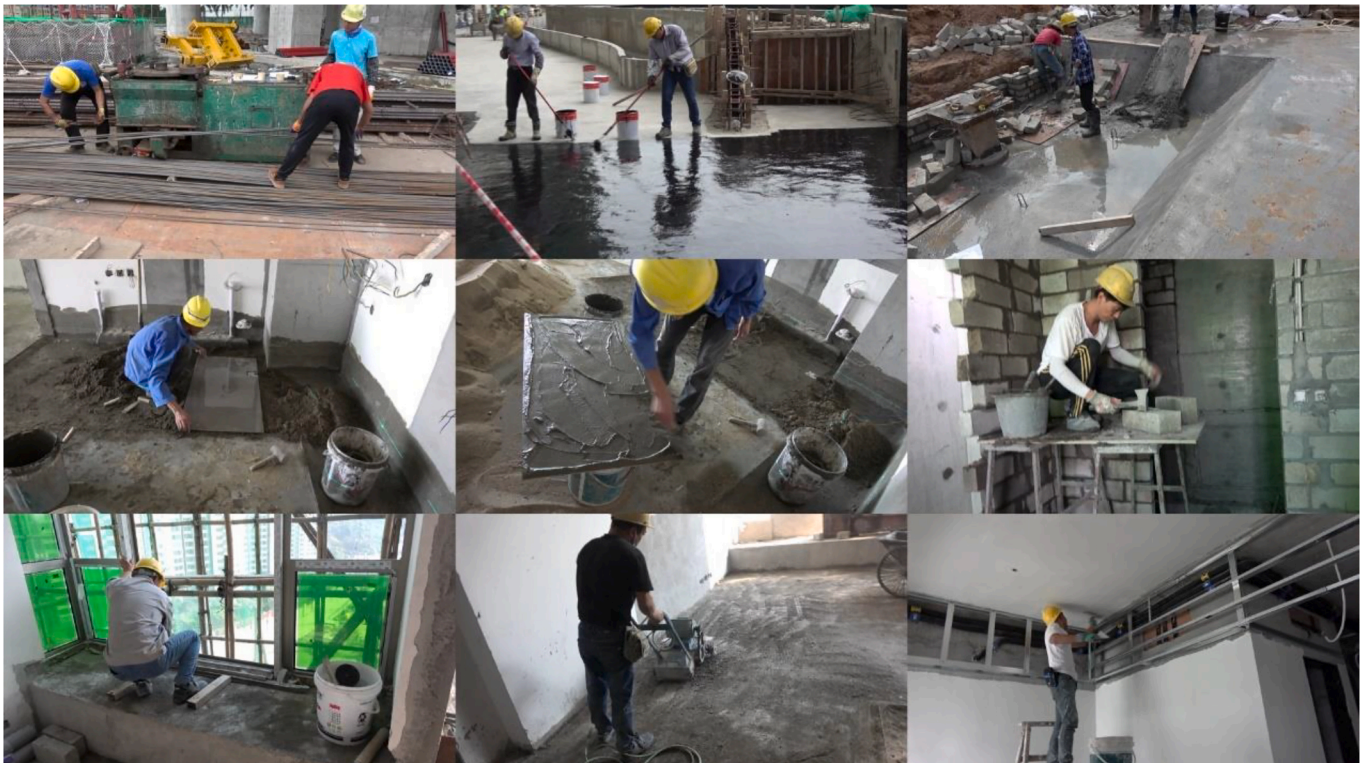


Fig. 10. Videos in the dataset.

Table 1
Comparison of ZSAR top-1 accuracy in forward and backward processes.

Methods\task	Seen	Unseen	Seen + Unseen
Forward	0.96	0.23	0.33
Backward	0.95	0.53	0.86

3.4. Overall process

The overall workflow of the framework is shown in Fig. 9. The solid black and dotted green lines indicate the forward process and the backward process, respectively. The forward process uses the VED module initialised by prior information to detect entities in the video. The backward process strengthens the prior information by distilling knowledge from general KGs and word embeddings. All potential KG entities fall into two categories: *Seen0* (in the training set) and *Unseen* (outside the training set), where *Unseen* = {*Unseen1*, *Unseen2*, *Unseen3*, ..., *UnseenN*}. The VED module is first initialised by the word embeddings of *Seen0* and *Unseen* (Prior1). Due to the noise in the word embeddings and the limited generalisation ability of *Seen0*, the VED module can only recognise a subset of *Unseen* (*Unseen1*). Therefore, *Unseen1* is converted into known data, namely, *Seen1* = {*Seen0*, *Unseen1*}, *Unseen* = {*Unseen2*, *Unseen3*, ..., *UnseenN*}. Next, the backward process performs, with Prior2 (semantic embeddings) replaces Prior1. Afterwards, the VED module, initialised by Prior2, detects entities from the video and obtains a new unseen subset (*Unseen2*). At this stage, *Seen2* = {*Seen0*, *Unseen1*, *Unseen2*}, *Unseen* = {*Unseen3*, ..., *UnseenN*}. As the two processes continue to iterate, the *Seen* set increases and the *Unseen* set decreases. The iteration stops when the forward process fails to recognise new entities.

4. Experiments

4.1. Experimental setup

4.1.1. Dataset description

As shown in Fig. 10, the ZSAR dataset contains 29 seen classes and 5 unseen classes. Each class contains an average of 3 to 5 videos, and each video is at least 1 min in length. The ZSD dataset contains 24 classes of tools or materials, 20 of which are seen, and 4 are unseen.

4.1.2. Evaluation metrics

In the ZSAR task, the top-1 accuracy, i.e., the accuracy with which the top-ranked category matches the actual results, is adopted as our evaluation metric. In the ZSD task, the mean average precision (mAP) is utilised to evaluate novel object detection performance, mainly because it has been widely used in supervised object detection.

4.1.3. Implementation details

A general KG ConceptNet 5.5 [52] is used for AEC KG updating. The word embeddings are obtained from the literature [54] using the word2vec model trained on the Baidu Encyclopaedia with a word embedding length of 300. The semantic branch for extracting semantic features utilises the GCN model with three graph convolution layers, and output channels are 2048, 1024 and 300, respectively. It is worth noting that each convolution layer contains all seen and unseen nodes, and the features of seen nodes and unseen nodes are leveraged in the training stage and testing stage, respectively. Only in the GZSL task do we use all the features, including seen and unseen.

4.2. Quantitative analysis

In this section, the top-1 accuracy of the forward and backward processes are compared under three different tasks, which are distinguished by test categories: Only Seen Class instances, Only Seen Class instances, Both Seen Class instances and Both Seen Class instances. Note

Table 2
Comparison of ZSD mAP in the forward and backward processes.

Methods\task	Seen	Unseen	Seen + Unseen
Forward	0.47	0.21	0.38
Backward	0.54	0.30	0.45

that we only perform one forward and one backward process and the training and testing datasets of these two processes are the same.

As shown in Table 1, although the top-1 accuracy of the forward process in the seen task is slightly better than that of the backward process, the figure of the forward process is much lower than those of the backward process regarding the Unseen task and Seen + Unseen task, which indicates that the backward process has better performance than the forward process.

Table 2 shows the comparison of the ZSD model between the forward and backward processes, and the mAP of these three tasks in the backward process is larger than that in the forward process. Tables 1 and 2 show that the backwards process's performance is better than that of the forward process, which means the backward process has constructed more robust prior information, and prior information is strengthened as the seen classes increase in the iterative process of the framework. Therefore, the feasibility of the video2entities framework is verified.

4.3. Qualitative results

4.3.1. Seen construction process recognition

The video of a seen construction process is shown in Fig. 11, with the label "ground levelling". After recognising the labels in the construction process through the forward and backward processes, 62 labels were identified. The correct rate of the backward process identification was 79.0%, and that of the forward process identification was 96.8%, as shown in Fig. 12.

4.3.2. Unseen class construction process identification

Fig. 13 shows an unseen construction process video with the label "window sill cleaning". Again, two processes were used to identify the construction process labels, as shown in Fig. 14, with 25 labels detected in total. The backward process's accuracy rate was 68.0%, and that of the forward process was 56%.

4.3.3. Seen construction tools and materials detection

As shown in Fig. 15, the forward process detects only the barrel, while the backward process detects more categories and obtains a higher recall rate.

4.3.4. Unseen construction tools and materials detection

As shown in Fig. 16, there are 4 types of materials and tools in the video, where barrel, tile and chainsaw are in the seen category, while nail belongs to the unseen category. The forward process detected tiles and barrels, but could not detect small objects such as chainsaws and nails, while the backward process detected all of these objects.

4.4. Discussion

4.4.1. More intuitive comparison between the forward and backward processes

Even though we have shown that the backward process outperforms the forward process in quantitative analysis, we are still looking for a more intuitive way to demonstrate the capability of VED to classify different categories. We visualise the feature maps of the last layer of the forward and backward processes using t-SNE [55], a dimensionality reduction algorithm for high-dimensional data. As shown in Fig. 17, different colours represent different categories, and each circle represents an instance. The backward process can cluster semantically similar instances together, whereas, in the forward process, these instances are



Fig. 11. Seen construction process video: ground levelling.

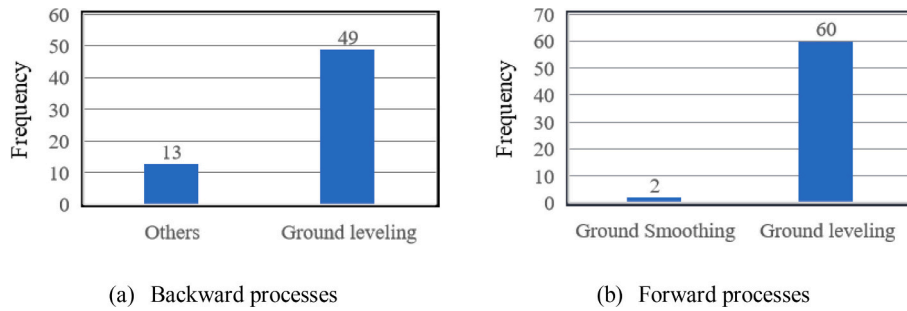


Fig. 12. Comparison of the results of seen construction process recognition between the two processes. Figures (a) and (b) are statistical histograms of the construction process labels.



Fig. 13. Unseen class construction process video: window sill cleaning.

mostly discrete, which means that the backward process has a better ability to separate different categories than the forward process.

4.4.2. Relationship graphs and prior information

Under three different evaluation indicators, the backward process has shown superior performance over the forward process, mainly due to the difference in prior information. Therefore, it is important to determine the effect of different prior information on model performance, if we construct prior information using different graph structures, e.g., a random graph and a fully connected graph. As shown in Tables 3 and 4, it is clear that the results of these two graphs are close to random guesses. It can be seen that a well-structured relationship graph may provide extra prior information that helps to improve the model performance, while a random graph has only negative effects.

4.4.3. Comparison of other methods

On the ZSAR task, we compared the forward and backward processes with the following methods: attribute-based, GAN [56], ConSE [33] and LATEM [57]. In the attribute-based method, we label the 40 attributes of the construction process by location, tools and materials. We also adapted GAN [56], ConSE [33] and LATEM [57] for ZSAR. The results shown in Table 5 indicate that the backward process obtained the best performance among all methods for both Unseen and Seen + Unseen tasks.

On the ZSD task, we adapted different ZSL methods based on YOLOv2 for comparisons, such as Zhu [29], YOLOv2 + ConSE [33] and YOLOv2 + polarity loss (PL) [58]. As shown in Table 6, on the Seen task, the forward process showed the best performance. The best method was

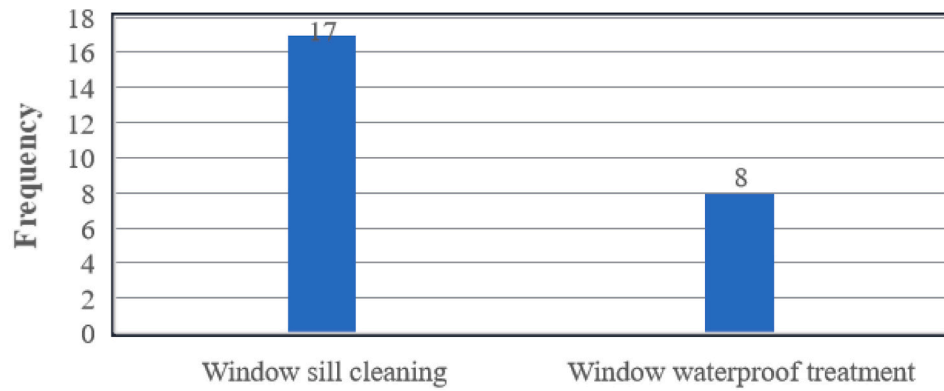
the backward process on the Unseen task, while it was Zhu [29] on the Seen + Unseen task.

4.4.4. The impact of training data on the model's generalisation capability

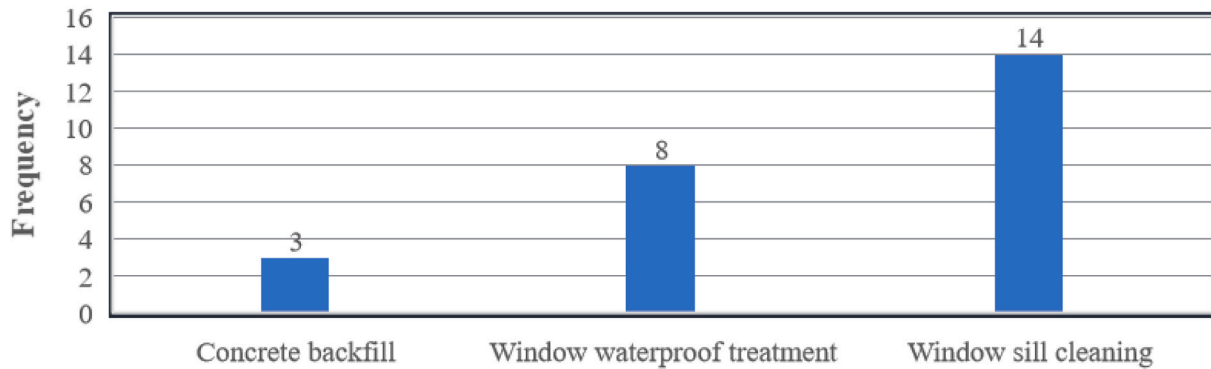
The core idea of ZSL is to generalise the known knowledge to the unknown knowledge. Due to limited training data and bias in the training dataset, the model's generalisation performance can be limited, so it is essential to quantify this limitation. We used different train/test splits to show the impact on the generalisation performance of the model. As shown in Tables 7 and 8, it is clear that the model performance on *Unseen* and *Seen + Unseen* tasks improves as the training data increases, implying that the increase in training data helps to improve the generalisation performance of the model.

5. Conclusions

In this study, we presented a computer vision-based video entity extraction framework for the AEC domain KG updates. We introduced the ZSL technique to develop two iterative processes (i.e., the forward and backward processes) for the extraction of unseen entities. Specifically, we initialised the VED using prior information to train the ZSD and ZSAR models for construction, process, tool and material detection. We also revealed that the backward process could strengthen the prior information, thus enhancing the performance of VED. To validate our approach, we tested our framework in architectural decoration videos. Experimental results showed that our framework could simultaneously extract seen and unseen entities and that the backward process achieved satisfactory results in both Seen and Seen + Unseen tasks compared with



(a) Backward processes



(b) Forward process

Fig. 14. Comparison of the results of unseen construction process recognition between the two processes. Figures (a) and (b) are the statistical histograms of the construction process labels.



(a) Forward process



(b) Backward process

Fig. 15. Test results of seen construction tools and materials.

existing methods. In conclusion, we can incrementally update the AEC domain KG by applying our framework to extract entities from the video and update them to the general KG.

It is worth mentioning that we still have considerable room for improvement in this study. For example, we have only extracted entities,

while the extraction of attributes and relationships is also critical to KG update. The current performance of ZSAR and ZSD models is not satisfactory. We did not test our framework on large-scale datasets. We will further address these issues in future work and focus on visual relationship extraction for AEC KG relationship updates.

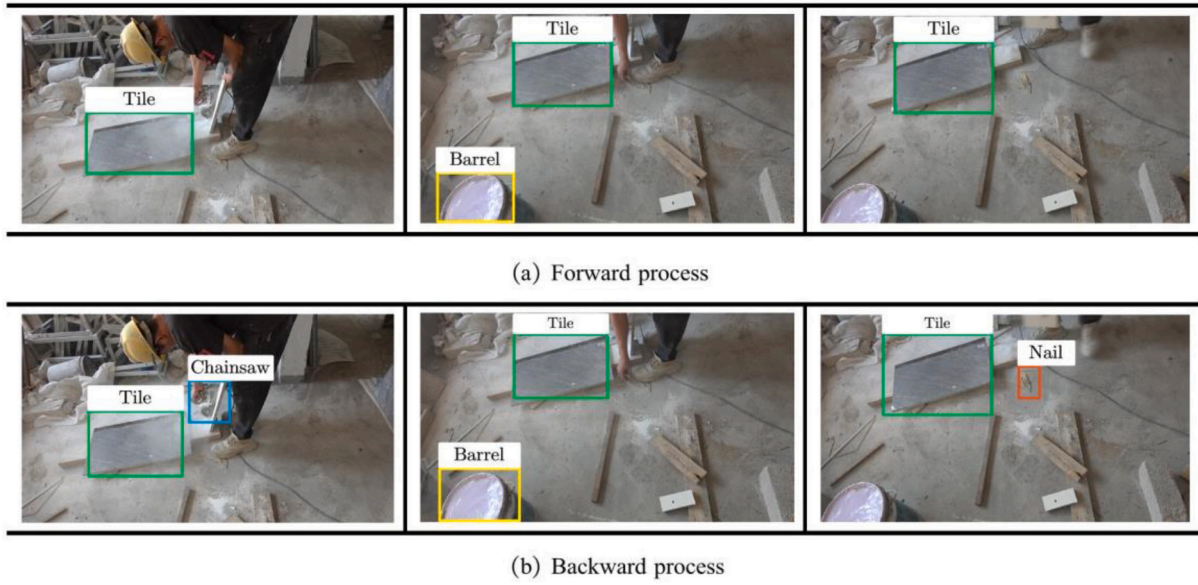


Fig. 16. Test results of unseen construction tools and materials.

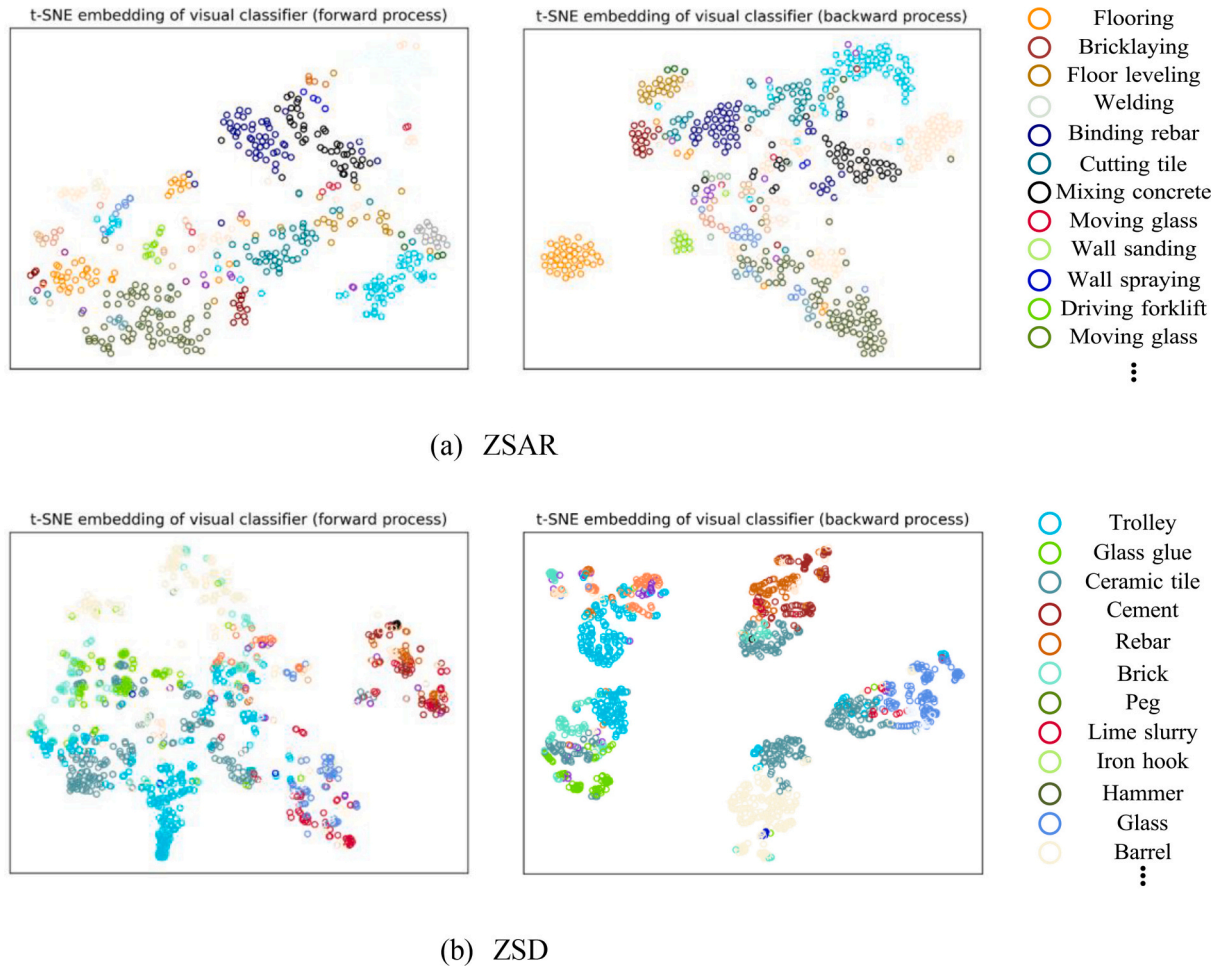


Fig. 17. t-SNE visualisations of the feature map of the last layer during the forward and backward processes.

Table 3

Comparison of ZSAR top-1 accuracy between different relationship graphs in the backward process.

Methods\task	Seen	Unseen	Seen + Unseen
Random	0.33	0.10	0.21
Fully connected	0.04	0.10	0.03

Table 4

mAP comparison of ZSD between different relationship graphs in the backward process.

Methods\task	Seen	Unseen	Seen + Unseen
Random	0.10	0.08	0.05
Fully Connected	0.04	0.09	0.02

Table 5

Comparison of ZSAR top-1 accuracy with the other methods.

Methods\task	Seen	Unseen	Seen + Unseen
Attributes	0.70	0.50	0.41
GAN	0.98	0.23	0.36
ConSE	0.97	0.50	0.81
LATEM	0.83	0.30	0.61
Forward	0.96	0.23	0.33
Backward	0.95	0.53	0.86

The bold font highlights the largest mAP of each task.

Table 6

Comparison of ZSD mAP with the other methods.

Methods\task	Seen	Unseen	Seen + Unseen
Zhu	0.70	0.28	0.49
YOLOv2 + ConSE	0.81	0.04	0.37
YOLOv2 + PL	0.71	0.11	0.42
Forward	0.82	0.11	0.54
Backward	0.54	0.30	0.45

The bold font highlights the largest mAP of each task.

Table 7

Comparison of ZSAR top-1 accuracy with different train/test splits in the backward process.

Train/test split \task	Seen	Unseen	Seen + Unseen
3:7	0.99	0.00	0.00
5:5	0.98	0.00	0.03
7:3	0.98	0.19	0.40
9:1	0.95	0.53	0.86

Table 8

mAP comparison of ZSD with different train/test splits in the backward process.

Train/test split \task	Seen	Unseen	Seen + Unseen
2:8	0.77	0.01	0.05
5:5	0.30	0.17	0.12
8:2	0.54	0.30	0.45

Declaration of Competing Interest

None.

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